Smart Electricity Billing for Multiple Apartments

Python Programming – Activity-Based Learning (ABL) Assignment

1. Problem Statement

Calculate apartment-wise electricity bills using progressive tariffs, fixed charges, and rebates; identify billing discrepancies and rank apartments by consumption.

2. Objectives

• Calculate consumption from meter readings. • Apply progressive tariff slabs. • Add fixed charges and deduct rebates. • Compare calculated and recorded bills. • Rank apartments by units consumed.

3. Assumptions and Tariff Policy

The first 100 units cost Rs. 2 per unit, the next 100 units cost Rs. 3 per unit, and units above 200 cost Rs. 5 per unit. A fixed charge of Rs. 100 is added. The specified rebate is deducted from the bill, with a minimum bill of zero. Recorded bills are compared against calculated bills to flag discrepancies.

4. Algorithm

```
1. Store apartment ID, previous/current meter readings, rebate, and recorded bill.
2. Calculate units = current reading - previous reading.
3. Calculate energy charge using progressive tariff slabs.
4. Add fixed charge and subtract rebate.
5. Compare calculated bill with recorded bill and flag mismatches.
6. Sort apartments by consumption in descending order.
7. Print apartment-wise bills, rankings, and billing summary.
```

5. Flowchart (Text Representation)

```
START
  |
Read apartment meter readings and billing records
  |
Calculate consumed units
  |
Apply progressive tariff + fixed charge - rebate
  |
Compare calculated bill with recorded bill
  |
Flag discrepancies
  |
Sort apartments by consumption (descending)
  |
Display bills, ranking, and summary
  |
 END
```

6. Python Source Code

```
# Smart Electricity Billing for Multiple Apartments
# Progressive tariffs, fixed charges, rebates, discrepancy checks, consumption ranking.

apartments = [
    {"id": "A101", "previous": 1200, "current": 1350, "rebate": 50, "recorded_bill": 1325},
    {"id": "A102", "previous": 800,  "current": 920,  "rebate": 0,  "recorded_bill": 790},
    {"id": "A103", "previous": 1500, "current": 1740, "rebate": 100, "recorded_bill": 2380},
    {"id": "A104", "previous": 600,  "current": 660,  "rebate": 25, "recorded_bill": 350},
]

FIXED_CHARGE = 100

def calculate_energy_charge(units):
    """Progressive tariff: first 100 units Rs. 2/unit,
    next 100 units Rs. 3/unit, above 200 units Rs. 5/unit."""
    if units <= 100:
        return units * 2
    elif units <= 200:
        return 100 * 2 + (units - 100) * 3
    else:
        return 100 * 2 + 100 * 3 + (units - 200) * 5

def calculate_bill(units, rebate):
    return max(0, calculate_energy_charge(units) + FIXED_CHARGE - rebate)

for apartment in apartments:
    apartment["units"] = apartment["current"] - apartment["previous"]
    if apartment["units"] < 0:
        apartment["units"] = None
        apartment["bill"] = None
```

```
        apartment["issue"] = "Invalid meter readings"
        continue

    apartment["energy_charge"] = calculate_energy_charge(apartment["units"])
    apartment["bill"] = calculate_bill(apartment["units"], apartment["rebate"])
    apartment["issue"] = (
        "Bill discrepancy"
        if apartment["recorded_bill"] != apartment["bill"]
        else "OK"
    )

valid_apartments = [a for a in apartments if a["units"] is not None]
ranked = sorted(valid_apartments, key=lambda a: (-a["units"], a["id"]))

print("SMART ELECTRICITY BILLING - MULTIPLE APARTMENTS")
print("=" * 100)
print("Progressive tariff: 0-100 units Rs.2/unit; 101-200 Rs.3/unit; above 200 Rs.5/unit")
print(f"Fixed charge: Rs.{FIXED_CHARGE} | Rebate is deducted from calculated bill")
print("-" * 100)
print(f"{'Apt ID':<9}{'Units':>8}{'Energy':>12}{'Fixed':>9}{'Rebate':>9}{'Calc Bill':>12}{'Recorded':>12}{'Status':>20}")
print("-" * 100)

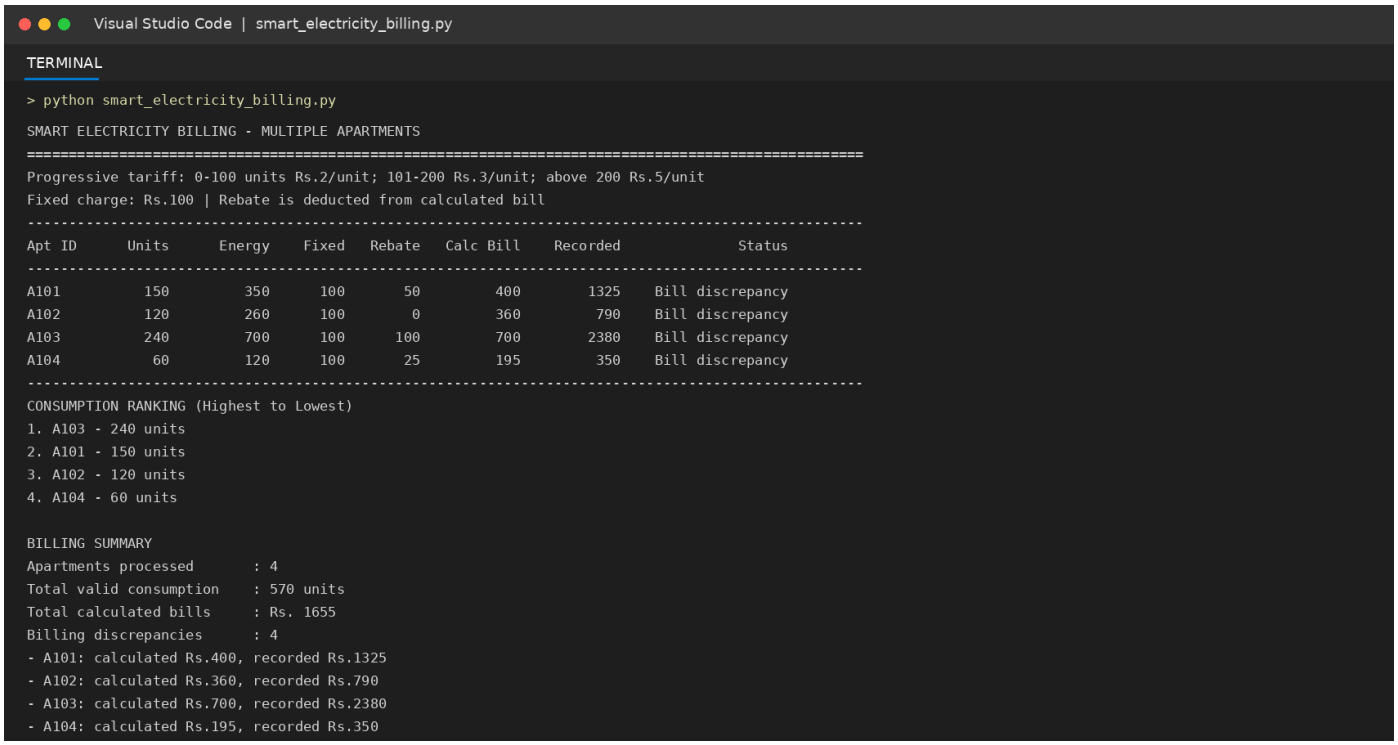
for a in apartments:
    if a["units"] is None:
        print(f"{a['id']:<9}{'N/A':>8}{'N/A':>12}{'N/A':>9}{a['rebate']:>9}{'N/A':>12}{a['recorded_bill']:>12}{a['issue']:>20}")
    else:
        print(f"{a['id']:<9}{a['units']:>8}{a['energy_charge']:>12}{FIXED_CHARGE:>9}"
              f"{a['rebate']:>9}{a['bill']:>12}{a['recorded_bill']:>12}{a['issue']:>20}")

print("-" * 100)
print("CONSUMPTION RANKING (Highest to Lowest)")
for rank, a in enumerate(ranked, start=1):
    print(f"{rank}. {a['id']} - {a['units']} units")

discrepancies = [a for a in valid_apartments if a["issue"] != "OK"]
print("\nBILLING SUMMARY")
print(f"Apartments processed       : {len(apartments)}")
print(f"Total valid consumption    : {sum(a['units'] for a in valid_apartments)} units")
print(f"Total calculated bills     : Rs. {sum(a['bill'] for a in valid_apartments)}")
print(f"Billing discrepancies      : {len(discrepancies)}")
for a in discrepancies:
    print(f"- {a['id']}: calculated Rs.{a['bill']}, recorded Rs.{a['recorded_bill']}")
print("Result: Billing calculations and consumption ranking completed.")
```

7. Sample Execution Output Screenshot

This terminal-style screenshot presents the sample output. Run smart_electricity_billing.py in Visual Studio Code to reproduce the results.



8. Test Cases

```
A101: 1350 - 1200 = 150 units. Energy Rs.350; fixed Rs.100; rebate Rs.50; bill Rs.400.
A102: 920 - 800 = 120 units. Energy Rs.260; fixed Rs.100; rebate Rs.0; bill Rs.360.
A103: 1740 - 1500 = 240 units. Energy Rs.700; fixed Rs.100; rebate Rs.100; bill Rs.700.
A104: 660 - 600 = 60 units. Energy Rs.120; fixed Rs.100; rebate Rs.25; bill Rs.195.
Expected ranking: A103, A101, A102, A104.
Recorded bills that differ from calculated bills are flagged.
```

9. Conclusion

The program calculates apartment-wise electricity bills with progressive tariffs, fixed charges, and rebates. It identifies discrepancies between recorded and calculated bills and ranks apartments by consumption.